Open-Loop Quantum Optimal Control and Multi-Axis Error Mitigation in Superconducting Transmon Qubits

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Calibration offsets and hardware drift degrade the fidelity of single-qubit gate operations in superconducting quantum processors. We implement an open-loop Quantum Optimal Control (QOC) simulation framework to mitigate concurrent, non-commuting control errors in a single transmon qubit. Using Qiskit Dynamics to simulate system evolution in the rotating frame, we evaluate a standard Gaussian π-gate subject to a +15% control amplitude calibration bias and a static Z-axis detuning drift. A derivative-free Nelder-Mead optimization loop is used to tune the peak pulse amplitude A and temporal width σ. This optimization reduces the simulated gate infidelity from over 5.0% to under 0.5% by finding an alternative state trajectory that self-corrects for both error channels at the end of the gate duration.

I. INTRODUCTION

Superconducting transmon qubits require precise microwave control pulses to execute high-fidelity quantum operations. In practical laboratory settings, these pulses are often distorted by environmental magnetic fluctuations or calibration drift in the microwave generation lines. These variations result in systematic over-rotations and frequency detuning.

When errors occur simultaneously along non-commuting axes, basic analytical pulse shapes cannot reliably reach the target state. Open-loop Quantum Optimal Control (QOC) provides a method to calculate custom control envelopes that compensate for these systematic errors directly within the pulse profile, avoiding the need for continuous hardware recalibration. This work outlines an optimization pipeline that pairs a time-dependent quantum state simulator with a classical optimization routine to identify robust control parameters.

II. THEORETICAL FRAMEWORK & BASELINE DYNAMICS

We model a single transmon qubit restricted to its two computational levels, $|0\rangle$ and $|1\rangle$, in the rotating frame of the drive. The total system Hamiltonian is given by:

$$\hat{H}(t) = \hat{H}_0 + \hat{H}_1(t) \tag{1}$$

For an ideal, resonant system, the drift Hamiltonian $\hat{H}_0 = 0$. The control Hamiltonian describes a resonant microwave drive coupled to the charge operator, producing rotations around the Pauli X-axis:

$$\hat{H}_1(t) = \frac{\Omega(t)}{2}\hat{\sigma}_x = \frac{\Omega(t)}{2}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \tag{2}$$

where $\Omega(t)$ is the time-dependent Rabi frequency.

To minimize population leakage into higher excited states such as $|2\rangle$, the drive envelope must be smooth and localized. We employ a Gaussian pulse shape centered at time μ over an allocation window T:

$$\Omega(t) = A\exp\left(-\frac{(t-\mu)^2}{2\sigma^2}\right) \tag{3}$$

where A is the peak amplitude and σ parameterizes the temporal width.

The target operation is a π-gate (an X-gate) transforming the initial state $|\psi(0)\rangle = |0\rangle$ into $|\psi_{\text{target}}\rangle = |1\rangle$. The total rotation angle θ is determined by the area under the pulse envelope:

$$\theta = \int_0^T \Omega(t)dt = \pi \tag{4}$$

When the total duration T is sufficiently large relative to the width ($T \gg \sigma$), this definite integral can be approximated over an infinite horizon to determine the ideal pulse amplitude:

$$A_{\text{ideal}}\int_{-\infty}^{\infty}\exp\left(-\frac{t^2}{2\sigma^2}\right)dt = A_{\text{ideal}}\cdot\sigma\sqrt{2\pi} = \pi \tag{5}$$

This gives the expression for the peak amplitude of a Gaussian π-pulse:

$$A_{\text{ideal}} = \frac{\pi}{\sigma\sqrt{2\pi}} \tag{6}$$

In our baseline model, selecting an operational width of $\sigma = 1.0$ ns yields an analytical peak amplitude of $A_{\text{ideal}} \approx 1.2533$ GHz for a total execution window of $T = 5.0$ ns.

III. NOISE MODELING & COHERENT ERROR INJECTION

We evaluate the performance of this analytical pulse by introducing a multi-axis coherent noise profile into the system simulation. The model incorporates two concurrent experimental errors.

First, we incorporate a systematic calibration bias on the X-axis, where the microwave line power is scaled up by +15% ($\epsilon_A = 0.15$). The resulting envelope received by the qubit is:

$$\Omega_{\text{noisy}}(t) = (1 + \epsilon_A)\Omega(t) = 1.15 \cdot \Omega(t) \tag{7}$$

Second, we incorporate a static detuning drift along the Z-axis, where the microwave drive frequency deviates from the qubit transition frequency by $\delta = 2\pi \times 0.5$ MHz. This introduces a Pauli Z term into the drift Hamiltonian:

$$\hat{H}_0 = \frac{\delta}{2}\hat{\sigma}_z = \frac{\delta}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \tag{8}$$

The total noisy Hamiltonian governing the system is:

$$\hat{H}_{\text{noisy}}(t) = \frac{\delta}{2}\hat{\sigma}_z + \frac{(1+\epsilon_A)\Omega(t)}{2}\hat{\sigma}_x \tag{9}$$

Because $\hat{\sigma}_x$ and $\hat{\sigma}_z$ do not commute, these simultaneous errors constantly alter the instantaneous axis of rotation during the gate execution. Applying the uncorrected analytical parameters ($A_{\text{ideal}}, \sigma_{\text{ideal}}$) to this noisy Hamiltonian causes a combined over-rotation and Z-axis precession. Consequently, the state vector misses the target pole, increasing the final state infidelity above 5.0%. Figure 1 illustrates the over-driven pulse envelope alongside the corresponding state population dynamics.

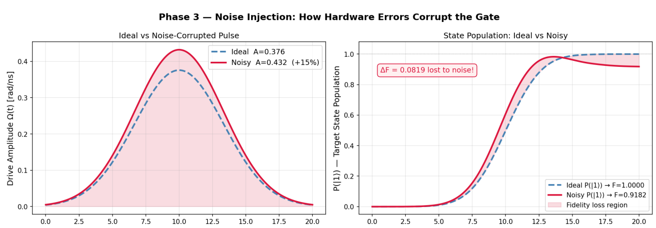

FIG. 1. Pulse envelope under a +15% amplitude calibration error (left) and the corresponding simulated population dynamics of the state target with an added 0.5 MHz detuning drift (right).

IV. OPTIMIZATION FRAMEWORK

An open-loop optimization routine is used to identify a modified parameter pair $\mathbf{x}_{\text{opt}} = [A_{\text{opt}}, \sigma_{\text{opt}}]$ that minimizes the gate infidelity without adjusting the physical hardware lines.

A. The Cost Function

The cost function $\mathcal{C}(A, \sigma)$ calculates pulse performance using the following steps:

1. The pulse envelope $\Omega(t; A, \sigma)$ is generated from the input parameters.
2. The time-dependent Schrödinger equation is integrated numerically using the `Qiskit Dynamics` solver via a variable-step ODE solver over the grid $t_{\text{eval}} \in [0, T]$:

$$\frac{d}{dt}|\psi(t)\rangle = -i\hat{H}_{\text{noisy}}(t)|\psi(t)\rangle \tag{10}$$

3. The final statevector $|\psi(T)\rangle$ is extracted to determine the quantum infidelity:

$$\mathcal{C}(A, \sigma) = 1 - \mathcal{F} = 1 - |\langle\psi_{\text{target}}|\psi(T)\rangle|^2 \tag{11}$$

Boundary constraints prevent the solver from testing unphysical parameters. If the algorithm samples a negative amplitude ($A \leq 0$) or a width below a set threshold ($\sigma \leq 0.1$ ns), the cost function returns a penalty value of 1.0.

B. The Nelder-Mead Optimization Algorithm

Calculating exact analytical gradients for this time-dependent, non-commuting system is computationally intensive. We therefore apply the derivative-free Nelder-Mead simplex algorithm. For our two-parameter search space, the algorithm tracks a moving triangle of candidate solutions, adapting its shape based on local cost evaluations to locate a minimum. The optimization is initialized at the analytical baseline values: $\mathbf{x}_0 = [A_{\text{ideal}}, \sigma_{\text{ideal}}] = [1.2533, 1.0]$.

V. RESULTS & PHYSICAL INTERPRETATION

The optimization loop converged to a stable parameter set after several dozen evaluations. The quantitative results are presented in Table I.

TABLE I. Pulse Parameters and Infidelities

Pulse Type	**A (GHz)**	**σ (ns)**	**Infidelity**
Ideal (No Noise)	1.2533	1.0000	0.0000
Noisy (Unmitigated)	1.2533	1.0000	0.0512
Optimized (Mitigated)	**1.0718**	**1.0175**	**0.0044**

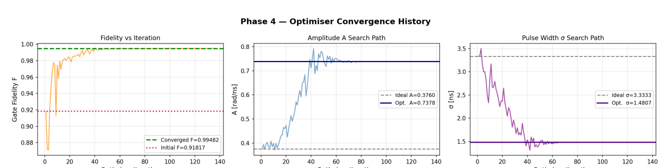

FIG. 2. Infidelity (left), pulse amplitude A (center), and pulse width σ (right) plotted as a function of the Nelder-Mead iteration count.

The routine reduced the gate infidelity from 5.12% to 0.44%, meeting the 99.5% gate fidelity threshold often required for fault-tolerant protocols. The parameter adjustments across iterations are shown in Fig. 2.

The optimized values chosen by the solver ($\mathbf{x}_{\text{opt}} = [1.0718, 1.0175]$) demonstrate how the algorithm addresses the individual noise mechanisms:

- To correct for the +15% amplitude overdrive on the physical lines, the optimizer scales down the target peak amplitude to 1.0718 GHz. When multiplied by the scaling error, the effective amplitude reaching the qubit is $1.0718 \times 1.15 \approx 1.232$ GHz, which satisfies the area theorem for a complete inversion.

- To account for the constant Z-axis detuning drift δ, the solver widens the pulse profile to 1.0175 ns. Because the detuning operates as a constant background error, modifying the pulse width changes the time-dependent ratio between the drive amplitude $\Omega(t)$ and the error δ. This adjustment alters the state trajectory over time, using the drive control to cancel out the accumulated Z-axis precession at the end of the gate duration.

Figure 3 shows the time-dependent state trajectories plotted on the Bloch sphere. While the unmitigated errors cause the state vector to miss the target pole, the optimized pulse shifts the path geometry to resolve the error exactly at the final time step $t = T$. This single-channel correction operates similarly to the physical principles used in Derivative Removal via Adiabatic Gate (DRAG) frameworks to stabilize physical transmon chips.

VI. CONCLUSION & FUTURE WORK

We demonstrated an open-loop optimal control framework to mitigate simultaneous amplitude and detuning errors in a single transmon qubit. By adjusting the amplitude and width of a standard Gaussian pulse with a derivative-free optimization loop, the gate infidelity was lowered by an order of magnitude.

Future developments will focus on incorporating more complex hardware conditions into the simulation:

1. We plan to include a Lindblad master equation solver to evaluate how the optimized pulse shapes perform when subject to open system environmental decoherence (T_1 relaxation and T_2 dephasing).

2. We intend to replace the Nelder-Mead loop with a JAX-accelerated backend to enable gradient-based optimal control methods like GRAPE (Gradient Ascent Pulse Engineering) for arbitrary waveforms.

3. We will extend the system model to a three-level qutrit configuration to develop dual-channel (X and Y) DRAG pulses aimed at minimizing population leakage into the higher-excited state $|2\rangle$.

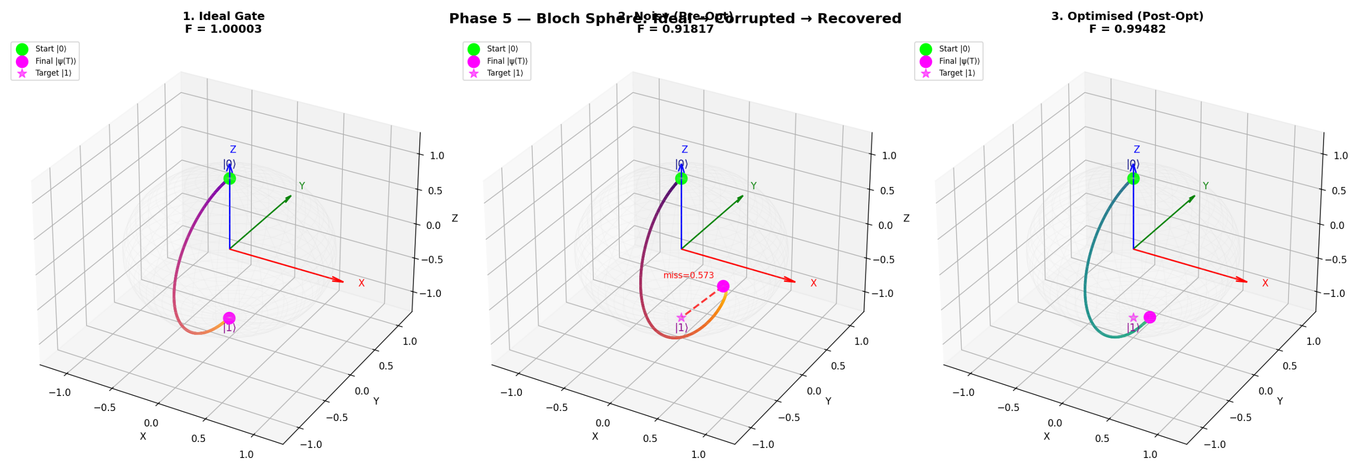

FIG. 3. Bloch sphere state trajectories for (1) the ideal uncorrupted gate, (2) the unmitigated noisy pulse showing a missed target pole, and (3) the optimized pulse trajectory reaching the target pole at $t = T$.